DR-APO: Decoupled-Representation Advantage-Guided Policy Optimization

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***Abstract*—Policy representations in reinforcement learning have expanded beyond Gaussian policies to diffusion and flow-matching based generative policies. However, existing RL methods remain representation-specific, lacking a unified optimization interface for emerging generative policies. In particular, extending likelihood-based objectives to multi-step generative policies couples optimization to their internal generation process, where accumulated likelihood ratios can weaken the update signal. Meanwhile, generative-policy-specific methods often rely on multi-candidate generation and critic-based selection, improving overall return while providing limited improvement to the policy itself. We introduce DR-APO (Decoupled-Representation Advantage-Guided Policy Optimization), which decouples policy improvement from policy representation through *advantage-guided target improvement* and *representation-compatible direct action fitting*, providing a shared optimization principle for Gaussian, diffusion, and flow policies. Across five MuJoCo-v5 tasks, DR-APO remains competitive with SAC for Gaussian policies and achieves the best aggregate policy-only performance for both diffusion and flow-matching policies. NavDP post-training improves the Success Rate (SR) from 70.33% to 89.55% in simulation and from 25% to 65% in real-world navigation. These results demonstrate a unified policy-improvement principle across Gaussian, diffusion, and flow policies, from continuous-control learning to embodied-policy post-training.**

I. Introduction

Mainstream RL policy-optimization methods are largely developed for specific policy representations, most notably Gaussian policies. Recent advances in diffusion and flow matching [1], [2] have enabled expressive generative policies, creating a growing need for RL methods that generalize across heterogeneous policy representations. However, existing optimization interfaces remain representation-specific and transfer poorly to multi-step generative policies.

One line of work extends likelihood-based optimization to generative policies. While PPO-style likelihood ratios are straightforward for Gaussian policies [3], diffusion and flow policies require optimization over their internal generation processes [4]–[7], coupling policy updates to multi-step generation and likelihood-ratio aggregation.

Another line exploits generative sampling to produce multiple candidates for critic-based evaluation and selection [8]–[10]. Although effective, execution performance can depend on inference-time critic selection. These limitations motivate a unified policy-optimization framework that improves the underlying policy across representations without explicit policy-likelihood evaluation or inference-time multi-candidate selection.

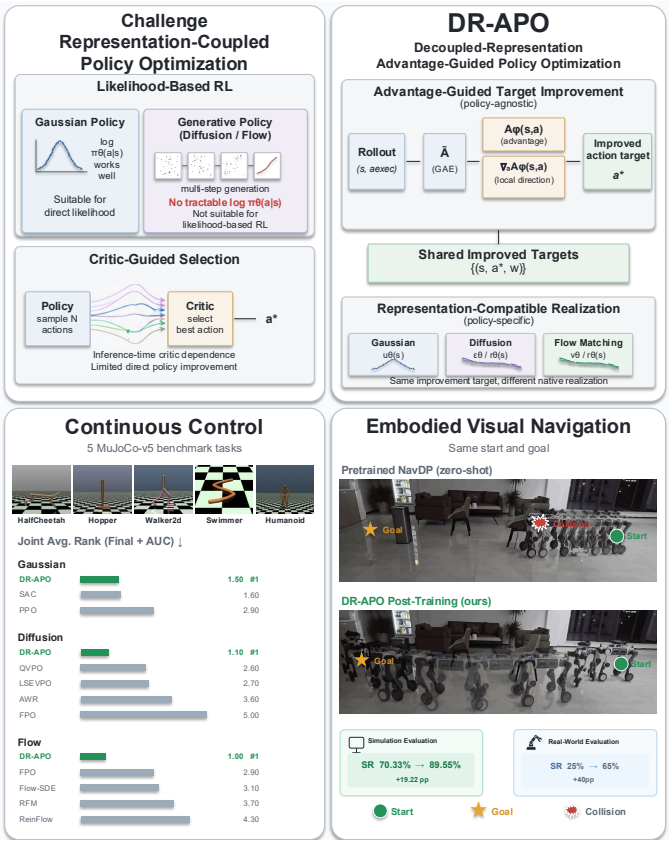


Fig. 1. Overview of DR-APO. Advantage-guided target improvement and representation-compatible direct action fitting enable unified optimization of Gaussian, diffusion, and flow policies across continuous control and embodied visual navigation.

We introduce **DR-APO (Decoupled-Representation Advantage-Guided Policy Optimization)**, a unified framework that decouples policy improvement from the underlying policy representation. DR-APO consists of *representation-decoupled policy optimization*, *advantage-*

guided target improvement, and *representation-compatible direct action fitting*. It first reformulates policy improvement as target optimization independent of policy parameterization, then constructs locally improved targets from advantage information, and finally realizes these targets through representation-compatible fitting. This enables a unified policy-improvement principle across Gaussian, diffusion, and flow policies without relying on likelihood-ratio actor updates or inference-time multi-candidate critic selection.

We evaluate DR-APO on continuous control and closed-loop embodied navigation. Across five MuJoCo-v5 tasks, DR-APO remains competitive with Gaussian policies, achieves the best result in $9/10$ diffusion comparisons, and ranks first in all $10/10$ flow comparisons, without inference-time multi-candidate critic selection. DR-APO further improves pretrained NavDP navigation success from 70.33% to 89.55% in simulation and from 25% to 65% on a DEEP Robotics M20 quadruped robot, a real-world gain of 40 percentage points.

Our contributions are threefold:

- We propose a representation-decoupled policy-optimization framework.
- We introduce advantage-guided target improvement and representation-compatible direct action fitting across Gaussian, diffusion, and flow policies.
- We validate DR-APO from MuJoCo continuous control to embodied-policy post-training.

II. Related Work

A. Conventional Policy Optimization

Continuous-control policy optimization has been extensively studied with parametric actors. TRPO and PPO optimize stochastic policies through trust-region or clipped likelihood-ratio objectives [3], [11], whereas DDPG and TD3 update deterministic actors through learned critic gradients [12], [13]. SAC combines off-policy value learning with maximum-entropy optimization [14], while MPO and AWR perform non-parametric improvement, projection, or advantage-weighted regression [15], [16]. Related weighted-fitting methods include RWR, AWAC, and CRR [17]–[19]. These paradigms are effective for conventional policies, but are generally developed around specific policy representations.

B. Generative Policy Optimization

Diffusion and flow models provide expressive multimodal policies for actions and trajectories [1], [2], [20]–[22]. Their iterative generation processes, however, make conventional optimization interfaces difficult to transfer directly. Existing methods mainly address this mismatch through internal-process policy optimization, value-guided learning or selection, and critic-driven action refinement.

One line adapts policy-gradient optimization to the generative process. DPPO treats diffusion denoising as an auxiliary decision process [4], while ReinFlow and π_{RL} introduce stochastic formulations for flow policies [6], [7]. FPO develops a PPO-style optimization interface for flow matching [23], while GenPO and ADPO further study on-policy or adaptive optimization of diffusion policies [24], [25]. These methods remain coupled to representation-specific denoising, flow dynamics, or surrogate objectives.

Another line incorporates learned value information into policy learning or execution. QVPO and RFM combine generative policies with critic-guided candidate selection [8], [10], while other approaches inject critic information through score matching, reverse diffusion, or value-reweighted objectives [5], [26], [27]. When multiple candidates are evaluated by a critic, execution performance additionally depends on inference-time selection.

Critic information can also refine actions before policy fitting. DIPO applies multi-step Q-gradient ascent before retraining a diffusion policy [28], while PA-RL combines critic-based reranking, local action optimization, and policy distillation [29]. Related flow-based methods transfer critic guidance into native vector-field objectives [30]. Repeated critic-gradient refinement can induce substantial target shifts, making the resulting supervision sensitive to critic estimation errors. Overall, existing approaches remain tied to representation-specific optimization, inference-time critic selection, or critic-driven action refinement. A shared policy-improvement mechanism across heterogeneous policy representations remains less explored.

C. Embodied Generative Policies and Post-Training

Generative policies are increasingly used in embodied control and navigation. Diffusion Policy applies diffusion models to visuomotor control [1], while NavDP generates multi-modal local trajectories for visual navigation [31]. Recent work further improves pretrained navigation policies through interaction, including reward-guided self-imitation [32], Q-reweighted post-training [33], online imitation [34], and GRPO-based fine-tuning [35]. These studies demonstrate the effectiveness of embodied-policy post-training and motivate evaluating whether a general policy-improvement framework can transfer from continuous control to pretrained embodied policies.

III. Method

A. Representation-Decoupled Policy Optimization

We consider a discounted Markov decision process (MDP) with state s_t and policy output a_t. Reinforcement learning seeks to maximize the expected discounted return:

$$\max_{\theta} J(\pi_\theta), \qquad J(\pi) = \mathbb{E}_\pi \left[\sum_{t=0}^{\infty} \gamma^t r_t \right], \tag{1}$$

where θ denotes the policy parameters, r_t is the reward received at decision step t, and $\gamma \in (0,1)$ is the discount factor.

Following the performance-difference lemma [36] and the first-order local surrogate used in TRPO [11], we approximate the policy update at iteration k as

$$\max_{\theta} \ \mathbb{E}_{s\sim\rho_k,\, a\sim\pi_\theta(\cdot|s)} \left[A^{\pi_k}(s,a) \right], \tag{2}$$

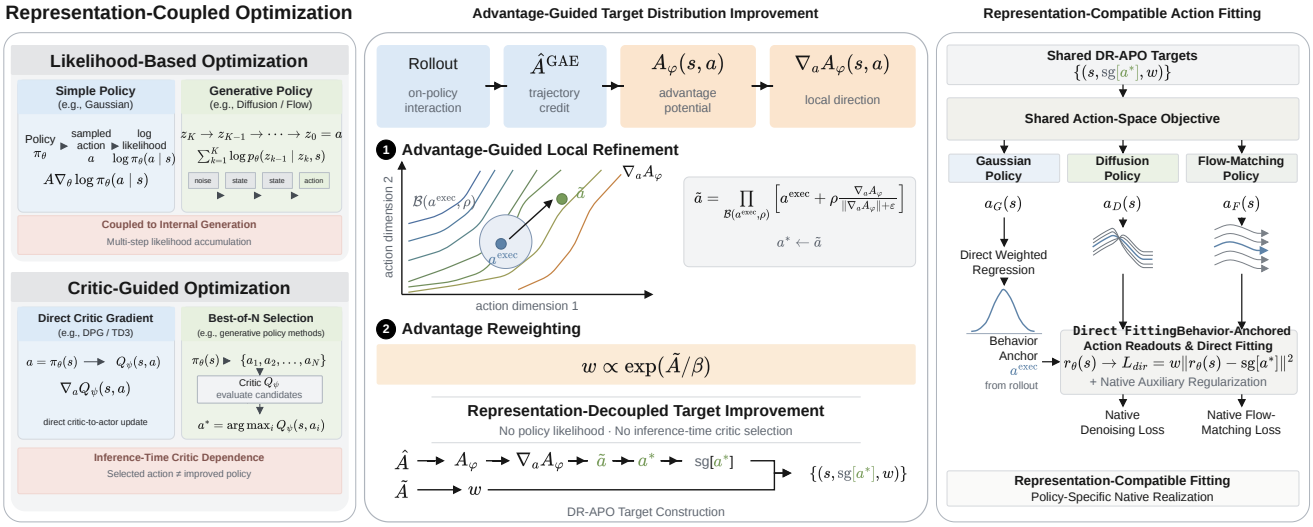


Fig. 2. Overview of DR-APO. Policy improvement is decoupled from policy representation through advantage-guided target improvement and representation-compatible direct action fitting across Gaussian, diffusion, and flow policies.

where $\pi_k = \pi_{\theta_k}$ is the current policy used to collect iteration-k rollouts, $A^{\pi_k}(s,a)$ is its advantage function, and ρ_k is its normalized discounted state-visitation distribution.

Introducing an auxiliary conditional action distribution $p(\cdot \mid s)$, we equivalently reformulate (2) as

$$\begin{aligned} \max_{p,\theta} \quad & \mathbb{E}_{s\sim\rho_k,\, a\sim p(\cdot|s)}\left[A^{\pi_k}(s,a)\right] \\ \text{s.t.} \quad & p(\cdot \mid s) = \pi_\theta(\cdot \mid s). \end{aligned} \tag{3}$$

Replacing this hard constraint with a distribution-consistency penalty yields the relaxed optimization problem

$$\begin{aligned} \max_{p,\theta} \Big\{ & \mathbb{E}_{s\sim\rho_k,\, a\sim p(\cdot|s)}\left[A^{\pi_k}(s,a)\right] \\ & - \lambda_{\mathrm{c}} \mathcal{D}_k(p,\pi_\theta) \Big\}, \end{aligned} \tag{4}$$

where $\lambda_{\mathrm{c}} > 0$ is the consistency penalty coefficient and $\mathcal{D}_k$ is a nonnegative discrepancy between the two conditional action distributions, averaged over states drawn from ρ_k.

We optimize (4) by coordinate ascent, alternating between the target distribution and policy parameters, analogous to the treatment in relative-entropy policy optimization [15]. At iteration k, we first fix the current policy $\pi_k = \pi_{\theta_k}$ and optimize with respect to p, obtaining

$$\begin{aligned} p_k^+ \in \arg\max_{p} \Big\{ & \mathbb{E}_{s\sim\rho_k,\, a\sim p(\cdot|s)}\left[A^{\pi_k}(s,a)\right] \\ & - \lambda_{\mathrm{c}} \mathcal{D}_k(p,\pi_k) \Big\}. \end{aligned} \tag{5}$$

Here, p_k^+ denotes the improvement target for the current iteration relative to π_k, rather than a globally optimal policy distribution.

With p_k^+ fixed, the expected-advantage term in (4) is independent of θ. The remaining optimization therefore reduces to minimizing the distribution-consistency error:

$$\theta_{k+1} \in \arg\min_{\theta} \mathcal{D}_k\left(p_k^+, \pi_\theta\right). \tag{6}$$

This decomposition yields two successive stages of policy optimization:

- **Representation-decoupled target improvement:** construct an improved target distribution p_k^+ from interaction feedback, independent of the policy's internal parameterization.
- **Representation-dependent target fitting:** fit p_k^+ through the learning interface of the selected policy representation.

Thus, the same improvement rule can be shared across Gaussian, diffusion, and flow-matching policies, while only the fitting interface is representation-specific.

B. Advantage-Guided Target Distribution Improvement

Equation (5) provides a general optimization problem for the target distribution p_k^+. We specify the distribution-consistency penalty as

$$\lambda_{\mathrm{c}} \mathcal{D}_k(p,\pi_k) = \mathbb{E}_{s\sim\rho_k} \inf_{q} \left[\beta D_{\mathrm{KL}}(q\|\pi_k) + \frac{1}{2\eta} W_2^2(q,p) \right], \tag{7}$$

where $p(\cdot \mid s)$ is the target action distribution being optimized, $q(\cdot \mid s)$ is an auxiliary action distribution, and $\beta, \eta > 0$ control probability redistribution and action displacement, respectively. Here, W_2 denotes the second-order Wasserstein distance. Substituting (7) into (5) gives, for each fixed state s,

$$\begin{aligned} \max_{p,q} \Big\{ & \mathbb{E}_{a'\sim p}[A^{\pi_k}(s,a')] - \frac{1}{2\eta} W_2^2(q,p) \\ & - \beta D_{\mathrm{KL}}(q\|\pi_k) \Big\}, \end{aligned} \tag{8}$$

where the conditioning on s is omitted from the distributions.

Using the optimal-coupling representation of W_2, we first optimize over p, obtaining

$$\begin{aligned}\Phi_{k,\eta}(s,a) &\in \arg\max_{a'\in\mathcal{A}}\left[A^{\pi_k}(s,a') - \frac{\|a'-a\|_2^2}{2\eta}\right],\\ A^+_{k,\eta}(s,a) &= A^{\pi_k}(s,\Phi_{k,\eta}(s,a))\\ &\quad - \frac{\|\Phi_{k,\eta}(s,a)-a\|_2^2}{2\eta}.\end{aligned} \tag{9}$$

Here, $\Phi_{k,\eta}$ is the ideal action-improvement map, and $A^+_{k,\eta}$ is the proximal improvement value after subtracting the displacement cost. Subsequent optimization over q yields

$$\begin{aligned}q_k^+(a\mid s) &= \frac{\pi_k(a\mid s)\exp\Big(A^+_{k,\eta}(s,a)/\beta\Big)}{Z_{k,\eta}(s)},\\ Z_{k,\eta}(s) &= \mathbb{E}_{a\sim\pi_k(\cdot|s)}\left[e^{A^+_{k,\eta}(s,a)/\beta}\right],\end{aligned} \tag{10}$$

where $Z_{k,\eta}(s)$ is the normalizing factor. The resulting target distribution is

$$\begin{aligned}p_k^+(\mathrm{d}a'\mid s) &= \int \delta_{\Phi_{k,\eta}(s,a)}(\mathrm{d}a')\\ &\qquad \times \frac{e^{A^+_{k,\eta}(s,a)/\beta}}{Z_{k,\eta}(s)}\,\pi_k(\mathrm{d}a\mid s),\end{aligned} \tag{11}$$

where δ_x denotes the Dirac measure at x. Equation (11) gives the ideal action-then-weight construction: first determine the improved action locations, then assign probability mass according to their proximal improvement values.

Finite-Sample Policy Improvement. A common class of policy-improvement methods approximates the target distribution by sampling K candidate actions at the same state and exponentially reweighting them according to their advantages. Let $a_j \overset{\text{i.i.d.}}{\sim} \pi_k(\cdot\mid s)$ and $A_j = A^{\pi_k}(s,a_j)$. Define

$$\begin{aligned}\omega_j &= \frac{\exp(A_j/\beta)}{\sum_{\ell=1}^K \exp(A_\ell/\beta)},\\ \Gamma_K^{\text{multi}}(s) &:= \mathbb{E}\left[\sum_{j=1}^K \omega_j A_j - \frac{1}{K}\sum_{j=1}^K A_j\right].\end{aligned} \tag{12}$$

Here, ω_j is the normalized weight of candidate j, and $\Gamma_K^{\text{multi}}(s)$ is the expected advantage gain over uniform candidate weighting, with expectation over the K candidate draws. Let $\sigma_A^2(s) := \mathrm{Var}_{a\sim\pi_k(\cdot|s)}[A^{\pi_k}(s,a)]$ denote the action-wise advantage variance. Under weak exponential tilting, $|A^{\pi_k}(s,a)|/\beta \ll 1$,

$$\Gamma_K^{\text{multi}}(s) \simeq \frac{K-1}{K\beta}\sigma_A^2(s). \tag{13}$$

DR-APO instead uses a single executed action $a\sim\pi_k(\cdot\mid s)$ and refines it using first-order advantage information. For a nonzero gradient, the idealized update and its expected advantage gain are

$$\begin{aligned}a^+ &= a + \rho\frac{\nabla_a A^{\pi_k}(s,a)}{\|\nabla_a A^{\pi_k}(s,a)\|_2},\\ \Gamma_1^{\text{ref}}(s) &:= \mathbb{E}_{a\sim\pi_k(\cdot|s)}\left[A^{\pi_k}(s,a^+) - A^{\pi_k}(s,a)\right].\end{aligned} \tag{14}$$

Here, a^+ is the refined action, $\rho>0$ is the refinement radius, and $\nabla_a A^{\pi_k}$ is the action gradient of the advantage. If $A^{\pi_k}(s,\cdot)$ has an L_A-Lipschitz gradient and the update remains feasible, then

$$\Gamma_1^{\text{ref}}(s) \ge \rho\,\mathbb{E}_{a\sim\pi_k(\cdot|s)}\left[\|\nabla_a A^{\pi_k}(s,a)\|_2\right] - \frac{L_A\rho^2}{2}.$$

Consequently, within the approximation in (13), the condition

$$\begin{aligned}&\mathbb{E}_{a\sim\pi_k(\cdot|s)}\left[\|\nabla_a A^{\pi_k}(s,a)\|_2\right]\\ &\qquad \ge \frac{K-1}{K\beta\rho}\sigma_A^2(s) + \frac{L_A\rho}{2}\end{aligned} \tag{15}$$

gives

$$\Gamma_1^{\text{ref}}(s) \ge \Gamma_K^{\text{multi}}(s).$$

Thus, under this approximation, a single executed action can provide an expected refinement gain no smaller than that of K-candidate exponential reweighting when its average advantage-gradient magnitude satisfies (15). This idealized comparison concerns exponential reweighting, not inference-time Best-of-N selection.

Practical Realization. Given a rollout batch $\mathcal{B}_k = \{(s_i, a_i^{\text{exec}}, \hat{A}_i)\}_{i=1}^B$, we fit a differentiable advantage estimator A_ϕ to the rollout advantage estimates $\hat{A}_i$ and compute its action gradient at each executed action:

$$\begin{aligned}\mathcal{L}_A(\phi) &= \frac{1}{B}\sum_{i=1}^B\left[A_\phi(s_i,a_i^{\text{exec}}) - \mathrm{sg}(\hat{A}_i)\right]^2,\\ g_i &= \nabla_a A_\phi(s_i,a)|_{a=a_i^{\text{exec}}},\end{aligned} \tag{16}$$

where sg denotes stop-gradient. The estimator A_ϕ supplies local action directions, while rollout advantages determine sample weights. We then construct a bounded refinement:

$$\begin{aligned}a_i^+ &= \Pi_{\mathcal{C}_i}\left[a_i^{\text{exec}} + \rho\frac{g_i}{\|g_i\|_2+\varepsilon}\right],\\ \mathcal{C}_i &= \mathcal{A}\cap\mathcal{B}_2(a_i^{\text{exec}},\rho).\end{aligned} \tag{17}$$

Here, $\mathcal{B}_2(a_i^{\text{exec}},\rho)$ is the Euclidean ball of radius ρ, $\Pi_{\mathcal{C}_i}$ denotes Euclidean projection, and $\varepsilon>0$ provides numerical stability. The resulting displacement $\delta_i := a_i^+ - a_i^{\text{exec}}$ satisfies $\|\delta_i\|_2 \le \rho$ by construction.

In practice, we use the rollout advantage grounded in the executed behavior as a reweighting proxy, rather than explicitly evaluating the proximal improvement value. With standardized advantages $\tilde{A}_i$, we obtain

$$\begin{aligned}w_i &= \frac{1}{C}\exp\left(\frac{\tilde{A}_i}{\beta_{\text{w}}}\right),\\ \mathcal{T}_k^+ &= \{(s_i,a_i^+,w_i)\}_{i=1}^B,\end{aligned} \tag{18}$$

where $\beta_{\text{w}}>0$ is the weighting temperature and $C>0$ is a common scaling constant. These weights specify relative fitting importance rather than per-state normalized probabilities. Equations (16)–(18) provide a finite-sample realization of the action-then-weight structure in (11), rather than an exact solution of the ideal distributional update. Different policy representations subsequently fit the same target set $\mathcal{T}_k^+$.

C. Representation-Compatible Direct Action Fitting

From Sec. III-B, we obtain

$$\mathcal{T}_k^+ = \{(s_i, a_i^+, w_i)\}_{i=1}^B, \quad a_i^+ = a_i^{\text{exec}} + \delta_i, \quad \|\delta_i\|_2 \le \rho. \tag{19}$$

The fitting stage seeks to realize these local refinements within a finite policy update. For Gaussian policies, we directly fit the policy mean to the refined action target, defining the action readout as

$$r_\theta^G(s_i) = \mu_\theta(s_i). \tag{20}$$

This readout is the distribution mean, rather than a sampled action.

Generative policies instead predict a diffusion noise field [37] or flow velocity field. A native-fitting baseline uses a_i^+ as the clean target:

$$\begin{aligned} \ell_{\text{nat}}^D &= \frac{1}{2}\left\|\epsilon_\theta(s_i, \alpha_\tau a_i^+ + \sigma_\tau \epsilon, \tau) - \epsilon\right\|_2^2, \\ \ell_{\text{nat}}^F &= \frac{1}{2}\left\|v_\theta(s_i, (1-\tau)a_i^+ + \tau\epsilon, \tau) - (\epsilon - a_i^+)\right\|_2^2. \end{aligned} \tag{21}$$

Although native fitting can learn the refined targets, it first translates δ_i into a representation-specific regression perturbation. Let $z_\tau^{\mathcal{R}}$ and $y_\tau^{\mathcal{R}}$ denote its input and target:

$$\begin{aligned} \Delta z_{i,\tau}^{\mathcal{R}} &= z_\tau^{\mathcal{R}}(a_i^+, \epsilon) - z_\tau^{\mathcal{R}}(a_i^{\text{exec}}, \epsilon), \\ \Delta y_{i,\tau}^{\mathcal{R}} &= y_\tau^{\mathcal{R}}(a_i^+, \epsilon) - y_\tau^{\mathcal{R}}(a_i^{\text{exec}}, \epsilon). \end{aligned} \tag{22}$$

For small refinements and negligible native residuals at the behavior inputs, the first-order change in the native-loss gradient is

$$\Delta g_i^{\mathcal{R}} \approx (J_\theta^{\mathcal{R}})^\top \left(J_z^{\mathcal{R}} \Delta z_{i,\tau}^{\mathcal{R}} - \Delta y_{i,\tau}^{\mathcal{R}}\right), \tag{23}$$

where $J_\theta^{\mathcal{R}}$ and $J_z^{\mathcal{R}}$ are predictor Jacobians evaluated at the current policy and behavior input. Specifically,

$$\begin{aligned} \Delta z_{i,\tau}^D &= \alpha_\tau \delta_i, & \Delta y_{i,\tau}^D &= 0, \\ \Delta z_{i,\tau}^F &= (1-\tau)\delta_i, & \Delta y_{i,\tau}^F &= -\delta_i, \end{aligned} \tag{24}$$

which gives

$$\begin{aligned} \Delta g_i^D &\approx \alpha_\tau (J_\theta^\epsilon)^\top J_z^\epsilon \delta_i, \\ \Delta g_i^F &\approx (J_\theta^v)^\top \left[(1-\tau)J_z^v + I\right]\delta_i. \end{aligned} \tag{25}$$

Thus, the refinement enters through field-level changes; the diffusion input perturbation is additionally scaled by α_τ.

A local field correction also propagates through the generator. For $x_{j-1} = F_j^\theta(x_j, f_j)$ and $a = x_0$, an isolated prediction change Δf_j has first-order endpoint effect

$$\Delta a^{(j)} \approx \left(\prod_{\ell=1}^{j-1} \frac{\partial F_\ell}{\partial x_\ell}\right) \frac{\partial F_j}{\partial f_j} \Delta f_j. \tag{26}$$

For K-step Euler flow with step size $1/K$ and state-transition Jacobian products bounded by C,

$$\|\Delta a^{(j)}\|_2 \le \frac{C}{K} \|\Delta v_j\|_2. \tag{27}$$

This describes a single-step perturbation, not the complete parameter update, which can affect multiple generation steps.

Behavior-Anchored Action Readouts. For generative policies, we construct differentiable clean-action estimates at behavior-anchored inputs. Auxiliary noise $\epsilon_{i,m}$ is introduced specifically for fitting:

$$\begin{aligned} z_{i,\tau_m}^D &= \alpha_{\tau_m} a_i^{\text{exec}} + \sigma_{\tau_m} \epsilon_{i,m}, \\ z_{i,\tau_m}^F &= (1-\tau_m) a_i^{\text{exec}} + \tau_m \epsilon_{i,m}. \end{aligned} \tag{28}$$

The corresponding diffusion and flow readouts are

$$r_\theta^D(s, z, \tau) = \frac{z - \sigma_\tau \epsilon_\theta(s, z, \tau)}{\alpha_\tau}, \tag{29}$$

$$r_\theta^F(s, z, \tau) = z - \tau v_\theta(s, z, \tau). \tag{30}$$

These follow from the native corruption or interpolation equations, recovering the clean action for an exact samplewise noise or velocity prediction. Such samplewise recovery is not assumed for learned predictors. We average M intermediate readouts:

$$r_\theta^{\mathcal{R}}(s_i, \xi_i) = \frac{1}{M}\sum_{m=1}^M r_\theta^{\mathcal{R}}(s_i, z_{i,\tau_m}^{\mathcal{R}}, \tau_m), \quad \mathcal{R} \in \{D, F\}, \tag{31}$$

where $\xi_i = \{(z_{i,\tau_m}^{\mathcal{R}}, \tau_m)\}_{m=1}^M$ is fixed during the corresponding policy update. These are inputs constructed from the behavior action, not states from its rollout sampling path. Unlike Eq. (21), the direct objective keeps these inputs anchored at a_i^{exec} while supervising the readouts with a_i^+.

Direct Action Fitting. All three policy families share the weighted action-space objective

$$\mathcal{L}_{\text{dir}}^{\mathcal{R}} = \frac{1}{B}\sum_{i=1}^B \frac{w_i}{2}\left\|r_\theta^{\mathcal{R}}(s_i, \xi_i) - \text{sg}(a_i^+)\right\|_2^2, \tag{32}$$

where $r_\theta^G(s_i, \xi_i) := \mu_\theta(s_i)$ is independent of ξ_i for Gaussian policies. Define $e_i^{\mathcal{R}} = r_{\theta_k}^{\mathcal{R}}(s_i, \xi_i) - a_i^{\text{exec}}$ and $J_i^r = \partial r_\theta^{\mathcal{R}}(s_i, \xi_i)/\partial\theta\big|_{\theta_k}$. The unweighted per-sample gradient is exactly

$$h_i^{\mathcal{R}} := \nabla_\theta \ell_{\text{dir}}^{\mathcal{R}}\big|_{\theta_k} = (J_i^r)^\top (e_i^{\mathcal{R}} - \delta_i). \tag{33}$$

Thus, δ_i enters the action-space residual explicitly. For Gaussian policies, $e_i^G = \mu_{\theta_k}(s_i) - a_i^{\text{exec}}$ is the mean-to-behavior-action offset; for generative policies, $e_i^{\mathcal{R}}$ is the behavior-readout reconstruction residual. Neither is assumed to vanish. For a smooth loss and sufficiently small η_θ,

$$\begin{aligned} \ell_{\text{dir}}^{\mathcal{R}}(\theta_k - \eta_\theta h_i^{\mathcal{R}}) = \ell_{\text{dir}}^{\mathcal{R}}(\theta_k) \\ - \eta_\theta \|h_i^{\mathcal{R}}\|_2^2 + \mathcal{O}(\eta_\theta^2). \end{aligned} \tag{34}$$

This local descent concerns the single-sample readout loss, not a guarantee of final-action displacement or return improvement. The weights w_i scale each target's contribution to the shared objective.

Since selected readouts do not constrain the complete generative field, we retain native training as an auxiliary regularizer:

$$\mathcal{L}^{\mathcal{R}} = \mathcal{L}_{\text{dir}}^{\mathcal{R}} + \lambda_{\text{nat}} \mathcal{L}_{\text{aux}}^{\mathcal{R}}, \qquad \mathcal{R} \in \{D, F\}. \tag{35}$$

The direct term supervises local action refinements, while the auxiliary term preserves the corresponding native field-learning signal. Gaussian policies use mean regression under the direct objective in Eq. (32).

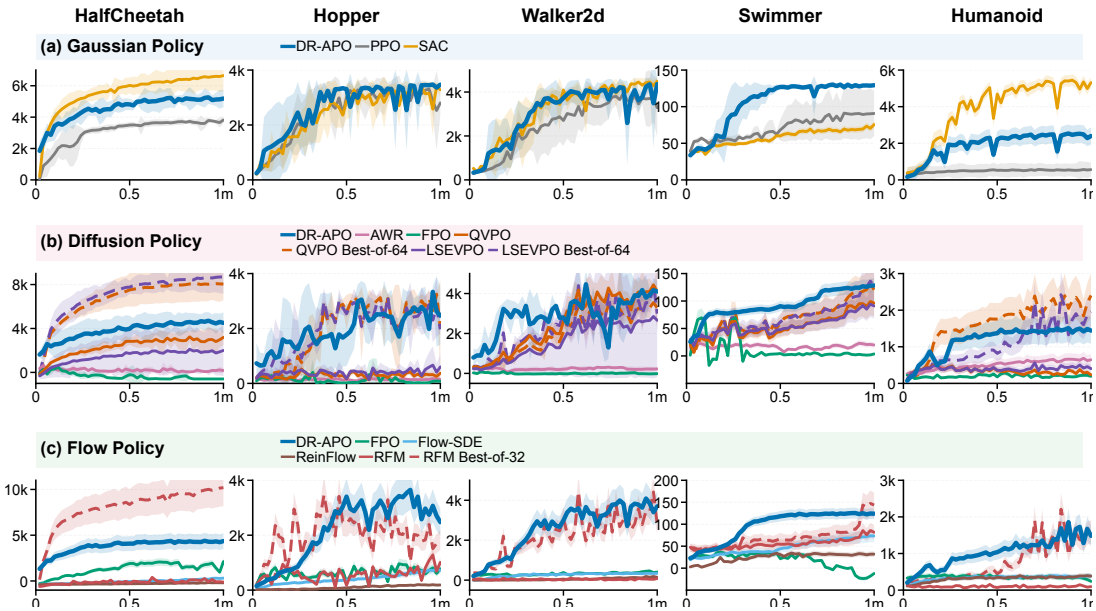


Fig. 3. Learning curves of Gaussian, diffusion, and flow policies on five MuJoCo-v5 tasks. Curves and shaded regions denote the mean and standard deviation.

IV. Experiments

We evaluate DR-APO across Gaussian, diffusion, and flow policies on five MuJoCo-v5 tasks, ablate its key components, and assess NavDP post-training in simulated and real-world navigation.

A. Experimental Setup

a) Continuous control: We train policies from scratch for 1M steps on five MuJoCo-v5 environments [38]: HalfCheetah (HC), Hopper (Hop), Walker2d (W2), Swimmer (Swi), and Humanoid (Hum). Policies are evaluated every 20k steps over five random seeds. Gaussian policies are compared with PPO and SAC; diffusion policies with AWR, FPO, QVPO, and LSEVPO; and flow policies with FPO, Flow-SDE, ReinFlow, and RFM. Best-of-64 variants of QVPO/LSEVPO and the Best-of-32 variant of RFM require additional critic-based selection, and are therefore reported separately and excluded from direct-policy rankings. We report **Final Return** over 820k–1M steps, normalized **AUC@1M** over 20k–1M steps, and the average rank across the five environments.

b) Embodied navigation: We post-train a pretrained diffusion-based NavDP policy with DR-APO using a wheeled robot in InternScenes [39]. Both simulation training and testing use this wheeled platform, with an unchanged execution controller; we report **SR** and **SPL**. Real-world trials use a DEEP Robotics M20 quadruped robot to compare pretrained NavDP with DR-APO post-training using SR (Table V).

B. Cross-Representation Policy Learning

Fig. 3 and Table I summarize the results across five MuJoCo-v5 environments. Under Gaussian policies, DR-APO achieves the best average AUC@1M rank while remaining competitive with SAC in Final Return. For diffusion policies, DR-APO achieves the highest mean AUC@1M on all five tasks and the highest mean Final Return on four, leading in 9 of the 10 environment–metric comparisons. For flow policies, DR-APO ranks first on both metrics across all five tasks.

These results show that the same target-level policy-improvement rule transfers across Gaussian, diffusion, and flow representations without requiring representation-specific likelihood optimization over their internal generation processes. Moreover, all DR-APO results are obtained by direct policy execution, without inference-time multi-candidate critic selection. Together, the results support representation-decoupled policy improvement while retaining strong within-family performance for both generative policy classes.

C. Ablation and Analysis

a) Comparison with Best-of-N: As shown in Table II, within the corresponding policy families, DR-APO improves direct-policy Final Return over LSEVPO and RFM on all five tasks, and over QVPO on four, without inference-time critic selection. For flow policies, DR-APO further outperforms RFM Best-of-32 on four of the five tasks, with HalfCheetah being the exception. For diffusion policies, the comparison is task dependent: DR-APO surpasses LSEVPO Best-of-64 on Hopper, Walker2d, and Swimmer, while trailing it on HalfCheetah and Humanoid.

b) Direct Action Fitting: We compare Native Only, Direct Only, and Full (Direct + Native Auxiliary). Native Only fits refined clean actions through the native diffusion/flow objectives, whereas Direct Only supervises behavior-anchored action readouts toward the same targets. As shown in Table III, Direct Only achieves higher mean Final Return than Native Only in both policy families, while Full obtains the highest mean Final Return among the three variants on all five tasks. These results support combining direct action fitting with native auxiliary training.

TABLE I

POLICY-ONLY RESULTS ON FIVE MUJOCO-V5 TASKS. WE REPORT FINAL RETURN, AUC@1M, AND WITHIN-FAMILY AVERAGE RANKS.

Policy	Method	HalfCheetah		Hopper		Walker2d		Swimmer		Humanoid		Avg. rank ↓	
		Final Return	**AUC@1M**	**Final Return**	**AUC@1M**	**Final Return**	**AUC@1M**	**Final Return**	**AUC@1M**	**Final Return**	**AUC@1M**	Final	AUC
Gaussian	PPO	3250.1	3057.5	3022.9	2484.4	3652.1	2547.9	68.4	53.4	541.0	493.2	3.00	2.80
	SAC	**6538.1**	**5621.8**	3158.3	2420.7	**4343.8**	3181.8	69.8	58.0	**5137.5**	**3951.5**	**1.40**	1.80
	DR-APO	5154.5	4574.1	**3288.9**	**2692.3**	4243.2	**3202.6**	**128.7**	**106.5**	2363.9	1945.0	1.60	**1.40**
Diffusion	AWR	135.5	205.7	144.9	143.9	220.2	228.1	19.2	17.3	640.6	509.2	3.60	3.60
	FPO	-587.5	-344.9	40.7	119.4	-2.0	-0.2	2.8	11.1	201.9	197.1	5.00	5.00
	QVPO	2981.3	2350.8	324.4	291.4	**3913.3**	2519.4	88.9	54.7	292.1	340.5	2.40	2.80
	LSEVPO	1924.6	1462.1	497.8	396.9	2521.9	1627.0	83.0	54.9	439.4	395.8	2.80	2.60
	DR-APO	**4572.7**	**3840.5**	**2631.5**	**1955.6**	3853.6	**2902.6**	**124.9**	**93.7**	**1462.0**	**1185.0**	**1.20**	**1.00**
Flow	FPO	1601.1	1311.8	736.2	619.0	390.5	314.3	-11.2	18.4	323.9	359.3	3.20	2.60
	Flow-SDE	215.7	-65.0	677.9	405.2	334.3	263.0	70.1	47.5	375.3	334.8	3.00	3.20
	ReinFlow	-211.8	-295.6	185.4	97.5	135.0	52.9	31.1	24.9	367.3	316.4	4.20	4.40
	RFM	46.0	-117.8	806.1	584.5	48.4	40.5	72.4	55.2	104.1	98.3	3.60	3.80
	DR-APO	**4299.2**	**3824.2**	**3016.4**	**2212.7**	**3650.0**	**2624.2**	**124.2**	**98.6**	**1535.4**	**1037.6**	**1.00**	**1.00**

TABLE II

DIRECT POLICY EXECUTION VERSUS BEST-OF-N CRITIC SELECTION.

	Diffusion Policy			*Flow Policy*	
Env.	QVPO P→B64	LSEVPO P→B64	**DR-APO** Direct	RFM P→B32	**DR-APO** Direct
HC	2981.3→8047.0	1924.6→8591.7	4572.7	46.0→9862.7	4299.2
Hop	324.4→2792.6	497.8→2506.8	2631.5	806.1→1839.8	3016.4
W2	3913.3→3130.7	2521.9→3316.4	3853.6	48.4→3211.1	3650.0
Swi	88.9→113.3	83.0→112.1	124.9	72.4→109.4	124.2
Hum	292.1→2191.2	439.4→1797.9	1462.0	104.1→1470.7	1535.4

P: direct policy execution; B64/B32: critic-selected Best-of-64/32.

TABLE III

ABLATION STUDIES ON TARGET IMPROVEMENT AND DIRECT ACTION FITTING. FINAL RETURN IS REPORTED AS MEAN ± SD OVER FIVE SEEDS.

Policy	Variant	HC	Hop	W2	Swi	Hum
(a) Target Improvement						
Gaussian	**Full**	**5155**±317	**3289**±263	**4243**±361	**128.7**±8.7	**2364**±423
	w/o Ref.	1531±263	1071±127	918±93	61.7±7.8	887±157
	w/o Weight.	3707±271	2361±216	2238±319	90.1±9.7	1857±383
Diffusion	**Full**	**4573**±391	**2632**±337	**3854**±431	**124.9**±9.8	**1462**±263
	w/o Ref.	1784±227	1289±301	1930±349	52.1±7.7	673±91
	w/o Weight.	3178±267	1945±359	2579±393	87.4±8.1	1086±169
Flow	**Full**	**4300**±409	**3016**±357	**3650**±451	**124.2**±9.1	**1535**±279
	w/o Ref.	1947±331	1291±241	1367±443	56.8±4.2	669±241
	w/o Weight.	3108±397	2413±281	2597±503	86.6±13.4	1172±269
(b) Direct Action Fitting						
Diffusion	**Full**	**4573**±391	**2632**±337	**3854**±431	**124.9**±9.8	**1462**±263
	Native Only	1132±121	576±99	667±83	31.2±4.9	467±71
	Direct Only	2961±357	1938±319	2728±363	90.2±8.3	1082±211
Flow	**Full**	**4300**±409	**3016**±357	**3650**±451	**124.2**±9.1	**1535**±279
	Native Only	973±431	826±339	871±91	29.9±10.8	376±219
	Direct Only	2878±367	2044±292	2599±414	89.8±7.2	1108±231

c) Target Improvement: We further ablate the two components of target improvement. *w/o Refinement* sets $a_i^+ = a_i^{\text{exec}}$ and retains only advantage weighting, whereas *w/o Weighting* sets $w_i = 1$ and retains only advantage-guided action refinement. As shown in Table III, removing either component degrades performance, indicating that advantage weighting and local action refinement provide complementary contributions to target improvement.

D. Embodied Navigation Post-Training

Table IV compares DR-APO with iPlanner [40], ViPlanner [41], and diffusion-based navigation baselines. DR-APO improves pretrained NavDP's overall SR from 70.33% to 89.55% (+19.22 percentage points), exceeding X-NavDP by 0.92 points. Its lower SPL (74.55 vs. 77.67) highlights a trade-off between navigation success and path efficiency. In real-world navigation, DR-APO further improves SR from 25% to 65%, a gain of 40 percentage points over pretrained NavDP (Table V). Figure 4 shows qualitative comparisons in simulation and the real world.

TABLE IV

CLOSED-LOOP WHEELED-ROBOT NAVIGATION PERFORMANCE ON INTERNSCENES. SR DENOTES SUCCESS RATE AND SPL DENOTES SUCCESS WEIGHTED BY PATH LENGTH.

Method	Overall		Commercial		Home	
	SR	SPL	SR	SPL	SR	SPL
iPlanner	45.28	43.54	51.75	50.02	38.80	37.05
ViPlanner	45.08	44.00	55.40	53.94	34.75	34.05
NavDP	70.33	66.85	72.55	69.74	68.10	63.96
NavOL	67.00	63.00	75.20	71.10	58.80	54.90
SIDP	77.43	69.32	82.15	74.16	72.70	64.48
NavDP-RL	–	–	–	–	68.70	62.80
X-NavDP	88.63	**77.67**	88.55	**78.96**	88.70	**76.37**
DR-APO	**89.55**	74.55	**89.30**	76.72	**89.80**	72.38

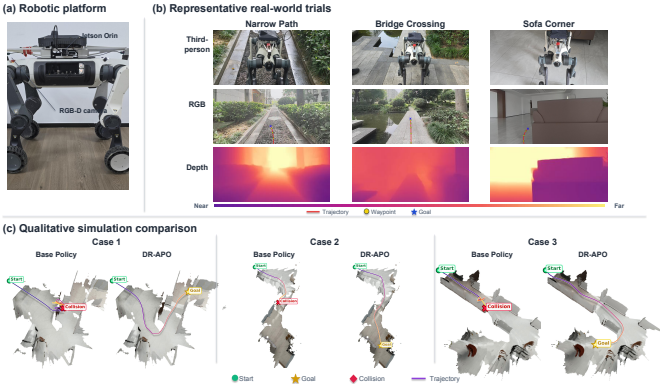


Fig. 4. Qualitative comparison of pretrained NavDP and DR-APO post-training in simulated and real-world navigation.

V. CONCLUSION

DR-APO decouples advantage-guided improvement from action fitting without likelihood-ratio updates or inference-time critic selection. On five MuJoCo-v5 tasks, it is competitive with Gaussian SAC and achieves the best aggregate policy-only results for diffusion and flow policies. NavDP post-training raises SR from 70.33% to 89.55% in simulation and from 25% to 65% in real-world navigation.

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TABLE V
REAL-WORLD NAVIGATION SUCCESS RATE (SR) ACROSS 20 SCENARIOS.

Method	NavDP (pretrained)	**DR-APO**
SR (%)	25	**65**

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